

Wentao Song

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I am a wildlife ecologist and applied statistician with comprehensive training in biology and ecology for wildlife management. My doctoral research focuses on modeling the spatiotemporal dynamics of North American breeding bird populations. I have contributed to a range of projects involving birds, rodents, deer, and primates. My interest is in applying statistical models and machine learning methods to analyze and predict population dynamics, with the goal of supporting wildlife management and conservation.

EDUCATION

Mississippi State University PhD. Natural Resources, Wildlife concentration Supervisor: Dr. Guiming Wang	Mississippi, US 2019 - 2024
Capital Normal University MSc. Zoology Supervisor: Dr. Zihui Zhang	Beijing, China 2015 - 2018
Institute of Zoology, Chinese Academy of Sciences Joint Master Program. Population ecology of small mammals Supervisor: Dr. Xinrong Wan	Beijing, China 2015 - 2018
China Agricultural University BSc(Agr.) Aquaculture	Beijing, China 2011 - 2015

EMPLOYMENT EXPERIENCE

Mississippi State University Teaching assistant Assisted instructors in preparing lectures and labs for undergraduate students.	Mississippi, US 2019.08-2024.12
Northwestern University Research assistant in Golden Snub-nosed monkey project Assisted the lab PI and postdoctoral researchers in coordinating government staff, stakeholders, and workers, as well as in data collection (tracking monkeys, monitoring behaviors, and 3D-reconstructing habitats), data analysis, and manuscript writing.	Xi'an, China 2018.09-2019.05

SKILLS

QUANTITATIVE SKILLS

- Time series analysis
- Spatial modeling
- Bayesian hierarchical modeling
- Structured decision making
- Coding with R, Python, Julia, and SAS
- Bayesian analysis with PyMC3, WinBUGS, JAGS, and Stan

TEACHING

- High school teaching certification (biology and physics) in China
- Ecology, Population Ecology, and Community Ecology

Ornithology and Mammalogy
Ecological Modeling

FIELDWORK & LABWORK

Small mammal trapping
Primate movement tracking
Deer spotlight survey and distance sampling
Bird banding
Polymerase Chain Reaction and Electrophoresis

LANGUAGES

Advanced speaker of Mandarin and English
Intermediate speaker of Japanese and French

COURSES ASSISTED

Applied Aquatic and Terrestrial Ecology (WFA3133, Dr. Sandra Correa). Fall 2023.
Mammalogy (WFA 4433, Dr. Christopher Ayers). Fall 2022.
Ornithology (WFA 4443, Dr. Christopher Ayers). Spring 2021.
Wildlife and Fishery Practice (WFA4473, Dr. John Davis). Spring 2020.

INVITED LECTURES

Responses of bird populations to climatic changes. Applied Ecology. MSU, Fall 2023.
Series Lectures of Conservation Biology:
(1)The IUCN red list of threatened species: How it's being used.
(2)Habitat destruction: Death by a thousand cuts.
(3)Reintroduction and rewilding as conservation strategies.
(4)Climate change: Implications for biodiversity.
Current Topics in Conservation Biology. MSU, Fall 2021.
Clustering Methods. Quantitative and Statistical Ecology. MSU, Spring 2021.
Plotting with Matplotlib. Python Club. MSU, Fall 2020.

CONFERENCE PRESENTATIONS

Variations in the responses of North American breeding bird populations to climatic changes. The Wildlife Society's 30th Annual Conference, Louisville, KY, 2023.
Variations in the responses of North American breeding bird populations to climatic changes. The Wildlife Society Mississippi Chapter Annual Meeting, Jackson, MS, 2023.
Population dynamic patterns of North American breeding birds under global changes. The Wildlife Society's 29th Annual Conference, Spokane, WA, 2022.
Machine Learning for Identifying Population Dynamic Patterns of North American Breeding Birds. The Wildlife Society Mississippi Chapter Annual Meeting (online), 2022.

Publications

Song, W., Kouba, A.J., Burger, L.M., Burger, L.W., Colvin, M.E., Wang, G. Interactions of life history traits, density dependence and responses to climate changes on long-term dynamic patterns of North American breeding bird populations. (In prep for Global Change Biology)
Shan, X., **Song, W.**, Strickland, B., Demarais, S., Beightol, C.V., Nelson, L., Wang, G. Population genetic clustering and fine-scale spatial genetic structure of white-tailed deer in Vicksburg National Military Park. (In prep)
Yang, H., Li, S., Hou, R., **Song, W.**, Fu, Y., Li, Y., ... & Li, B. (2023). Three-dimensional assessment of movement patterns of Sichuan snub-nosed monkeys affected by habitat structure in temperate forests. *Zoological Research*, 44(2), 361.
Song, W., Wang, Y., Zhang, X., Zhang, W., ... & Wan, X. (2017). Influence of Group Size and Foraging Distance on Vigilance Frequency of Brandt's Vole (*Lasiopodomys brandtii*) in Food Storing Period. *Chinese Journal of Zoology*, 52(5): 754-760.

Song, W., Wang, Y., Sai, N., ... & Zhang, Z. (2016). Numerical Response of Hawks Density to the Rodents Density in Typical Steppe. Chinese Journal of Zoology, 51(4): 529-535.

REFEREES

Dr. Guiming Wang (PhD advisor)

Professor, Mississippi State University

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Dr. Xinrong Wan (Master's advisor)

Associate Professor, Institute of Zoology, Chinese Academy of Sciences

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Dr. Xueyan Shan (Project advisor)

Associate Research Professor, Mississippi State University

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Dr. Sandra Correa (TA advisor)

Associate Professor, Mississippi State University

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